

THINNESS OF C-32

ABSTRACT. We prove that the degree-six C-32 hypergeometric monodromy group is thin. After an explicit rational change of basis, the group is generated by a signed cyclic shift and a rank-one transvection. The proof is a projective ping-pong argument on the projectivization of an explicitly given rational polyhedral cone and its reflected image. The only computer-assisted step is a finite certificate consisting of rational cone-containment checks, all encoded by an integral matrix displayed in the paper. The resulting abstract group is

$$(\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) *_{\mathbb{Z}/2\mathbb{Z}} \mathbb{Z}/12\mathbb{Z}.$$

Consequently the group is Zariski dense in its ambient symplectic group but has infinite index in the corresponding arithmetic lattice.

1. THE GROUP

Consider the hypergeometric parameters

$$\alpha = (0, 0, 0, 0, 1/6, 5/6), \quad \beta = (1/4, 3/4, 1/12, 5/12, 7/12, 11/12).$$

Equivalently,

$$f(x) = (x-1)^4 \Phi_6(x) = x^6 - 5x^5 + 11x^4 - 14x^3 + 11x^2 - 5x + 1,$$

and

$$g(x) = \Phi_4(x) \Phi_{12}(x) = x^6 + 1.$$

Using the low-degree-first companion convention, let

$$A_0 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 & 0 & -11 \\ 0 & 0 & 1 & 0 & 0 & 14 \\ 0 & 0 & 0 & 1 & 0 & -11 \\ 0 & 0 & 0 & 0 & 1 & 5 \end{pmatrix}, \quad B_0 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Let

$$\Gamma = \langle A_0, B_0 \rangle \leq \mathrm{GL}_6(\mathbb{Z}).$$

We shall work instead with

$$T_0 = B_0 A_0^{-1}.$$

Since $A_0 = T_0^{-1} B_0$, one has

$$\Gamma = \langle T_0, B_0 \rangle.$$

A direct calculation gives

$$T_0 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 5 & 1 & 0 & 0 & 0 & 0 \\ -11 & 0 & 1 & 0 & 0 & 0 \\ 14 & 0 & 0 & 1 & 0 & 0 \\ -11 & 0 & 0 & 0 & 1 & 0 \\ 5 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

The matrices A_0 and B_0 preserve the alternating form

$$\Omega_0 = \begin{pmatrix} 0 & 13 & 22 & 25 & 22 & 13 \\ -13 & 0 & 13 & 22 & 25 & 22 \\ -22 & -13 & 0 & 13 & 22 & 25 \\ -25 & -22 & -13 & 0 & 13 & 22 \\ -22 & -25 & -22 & -13 & 0 & 13 \\ -13 & -22 & -25 & -22 & -13 & 0 \end{pmatrix}.$$

Indeed

$$A_0^t \Omega_0 A_0 = \Omega_0, \quad B_0^t \Omega_0 B_0 = \Omega_0,$$

and

$$\det(\Omega_0) = 64 \neq 0.$$

Thus

$$\Gamma \leq \mathrm{Sp}_{\Omega_0}(\mathbb{Z}).$$

We now put the pair (B_0, T_0) into a normal form. Let

$$v = (0, 5, -11, 14, -11, 5)^t = (T_0 - I)e_0,$$

and define

$$P = [v, -B_0v, B_0^2v, -B_0^3v, B_0^4v, -B_0^5v].$$

Explicitly,

$$P = \begin{pmatrix} 0 & 5 & 11 & 14 & 11 & 5 \\ 5 & 0 & -5 & -11 & -14 & -11 \\ -11 & -5 & 0 & 5 & 11 & 14 \\ 14 & 11 & 5 & 0 & -5 & -11 \\ -11 & -14 & -11 & -5 & 0 & 5 \\ 5 & 11 & 14 & 11 & 5 & 0 \end{pmatrix}, \quad \det(P) = -64.$$

Conjugating by P , set

$$S = P^{-1}B_0P, \quad T = P^{-1}T_0P.$$

Then

$$S = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix},$$

so

$$Se_i = -e_{i+1} \quad (0 \leq i \leq 4), \quad Se_5 = e_0,$$

and

$$S^6 = -I, \quad S^{12} = I.$$

Moreover

$$T = \begin{pmatrix} 1 & 5 & 11 & 14 & 11 & 5 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Thus T is the transvection

$$T(y_0, \dots, y_5) = (y_0 + 5y_1 + 11y_2 + 14y_3 + 11y_4 + 5y_5, y_1, \dots, y_5).$$

In this basis the symplectic form is $4Q$, where

$$Q = \begin{pmatrix} 0 & 5 & 11 & 14 & 11 & 5 \\ -5 & 0 & 5 & 11 & 14 & 11 \\ -11 & -5 & 0 & 5 & 11 & 14 \\ -14 & -11 & -5 & 0 & 5 & 11 \\ -11 & -14 & -11 & -5 & 0 & 5 \\ -5 & -11 & -14 & -11 & -5 & 0 \end{pmatrix}.$$

Equivalently,

$$P^t \Omega_0 P = 4Q.$$

One checks

$$S^t Q S = Q, \quad T^t Q T = Q,$$

and

$$\det(Q) = 64.$$

Hence the conjugate group

$$\Gamma' = \langle S, T \rangle$$

lies in $\mathrm{Sp}_Q(\mathbb{Z})$. Since Γ and Γ' are rationally conjugate, proving thinness for Γ' proves thinness for Γ .

We shall also use the involution

$$E = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

It satisfies

$$E^2 = I, \quad ESE = S^{-1}, \quad ETE = T^{-1}.$$

2. ZARISKI DENSITY

Let $N = T - I$. Since $N^2 = 0$, the one-parameter subgroup

$$u \mapsto I + uN$$

lies in the Zariski closure of Γ' . Therefore the Lie algebra of the Zariski closure contains N , and hence also the conjugates

$$N_i = S^i N S^{-i}, \quad i = 0, \dots, 5.$$

Let $\mathfrak{l}$ be the Lie subalgebra of $\mathfrak{gl}_6(\mathbb{Q})$ generated by these six matrices.

$$\mathfrak{l} = \mathfrak{sp}_Q.$$

Proof. Every N_i lies in $\mathfrak{sp}_Q$, so $\mathfrak{l} \subseteq \mathfrak{sp}_Q$. The Lie algebra $\mathfrak{l}$ is obtained by repeatedly adjoining commutators

$$[X, Y] = XY - YX$$

and row-reducing the resulting matrices over $\mathbb{Q}$. The exact computation gives

$$\dim_{\mathbb{Q}} \mathfrak{l} = 21.$$

Since $\dim \mathfrak{sp}_6 = 3(2 \cdot 3 + 1) = 21$, it follows that

$$\mathfrak{l} = \mathfrak{sp}_Q.$$

The verification is included in Appendix B; it uses only rational arithmetic. $\square$

The group Γ' , and hence also Γ , is Zariski dense in its ambient symplectic group.

3. THE CONE CERTIFICATE

We work on real projective space $\mathbb{P}^5(\mathbb{R})$. For $i = 0, \dots, 5$, define the projective dominance chamber

$$\Delta_i = \{[y] \in \mathbb{P}^5(\mathbb{R}) : |y_i| > |y_j| \text{ for all } j \neq i\}.$$

Because S is a signed cyclic shift,

$$S\Delta_i = \Delta_{i+1},$$

with indices taken modulo 6. The chambers

$$\Delta_0, \dots, \Delta_5$$

are pairwise disjoint.

Let H be the 77×6 integer matrix whose rows are displayed in Appendix A. Define the closed rational polyhedral cone

$$K = \{y \in \mathbb{R}^6 : Hy \geq 0\}.$$

Let K° denote its topological interior.

The first key point is dominance.

$$\mathbb{P}(K^\circ) \subseteq \Delta_0.$$

Proof. Among the rows of H are the ten inequalities

$$y_0 - y_i \geq 0, \quad y_0 + y_i \geq 0, \quad 1 \leq i \leq 5.$$

Hence every $y \in K$ satisfies

$$y_0 \geq |y_i|, \quad 1 \leq i \leq 5.$$

The cone K is full-dimensional, because

$$z = (47, -1, -1, -1, -1, -1)^t$$

satisfies

$$Hz > 0.$$

Therefore every point of K° satisfies the displayed inequalities strictly:

$$y_0 > |y_i|, \quad 1 \leq i \leq 5.$$

Thus $[y] \in \Delta_0$. □

Now define eleven branch maps. Let

$$G_0 = T^{-1}.$$

For $1 \leq k \leq 5$, put

$$\varepsilon_k = (-1)^{k+1},$$

and define

$$G_{2k-1} = \varepsilon_k T^{-1} S^k, \quad G_{2k} = \varepsilon_k T^{-1} S^k E.$$

Explicitly, these are

$$T^{-1},$$

together with

$$\begin{aligned} & T^{-1}S, \quad T^{-1}SE, \\ & -T^{-1}S^2, \quad -T^{-1}S^2E, \\ & T^{-1}S^3, \quad T^{-1}S^3E, \\ & -T^{-1}S^4, \quad -T^{-1}S^4E, \end{aligned}$$

and

$$T^{-1}S^5, \quad T^{-1}S^5E.$$

The signs are projectively irrelevant; they are chosen only so that the cone K maps into the same affine chart.

[Finite cone certificate] For every $i = 0, \dots, 10$,

$$G_i K \subseteq K.$$

Proof. For a closed polyhedral cone

$$K = \{y : Hy \geq 0\},$$

the inclusion $GK \subseteq K$ is equivalent to the following dual condition: for every row h of H , the row hG belongs to the nonnegative rational cone generated by the rows of H . Thus it is enough to check

$$hG_i \in \text{Cone}_{\mathbb{Q}_{\geq 0}}\{H_1, \dots, H_{77}\}$$

for each row h of H and each $i = 0, \dots, 10$.

This gives $77 \cdot 11 = 847$ exact rational cone-membership checks in $\mathbb{Q}^6$. The verification script in Appendix B performs exactly these checks over $\mathbb{Q}$. No floating-point arithmetic is used. □

Since each G_i is invertible and K is full-dimensional, the inclusion

$$G_i K \subseteq K$$

implies

$$G_i K^\circ \subseteq K^\circ.$$

4. PING-PONG

Let

$$X^+ = \mathbb{P}(K^\circ), \quad X^- = EX^+, \quad X = X^+ \cup X^-.$$

Set

$$Y = \bigcup_{k=1}^5 S^k X.$$

Since

$$X^+ \subseteq \Delta_0$$

and $E\Delta_0 = \Delta_0$, we have

$$X \subseteq \Delta_0.$$

Therefore

$$S^k X \subseteq \Delta_k, \quad 1 \leq k \leq 5.$$

The chambers $\Delta_0, \dots, \Delta_5$ are pairwise disjoint, so

$$X \cap Y = \emptyset.$$

We next verify the ping-pong dynamics. From $G_0 K^\circ \subseteq K^\circ$, we get

$$T^{-1} X^+ \subseteq X^+.$$

For $1 \leq k \leq 5$, the inclusion

$$G_{2k-1} K^\circ \subseteq K^\circ$$

gives, projectively,

$$T^{-1} S^k X^+ \subseteq X^+.$$

Similarly,

$$G_{2k} K^\circ \subseteq K^\circ$$

gives

$$T^{-1} S^k EX^+ \subseteq X^+.$$

Since $X^- = EX^+$, this says

$$T^{-1} S^k X^- \subseteq X^+.$$

Hence

$$T^{-1}(Y \cup X^+) \subseteq X^+.$$

Conjugating by E , and using

$$ETE = T^{-1}, \quad ESE = S^{-1},$$

we obtain

$$T(Y \cup X^-) \subseteq X^-.$$

It follows by induction that

$$T^m Y \subseteq X \quad \text{for every } m \in \mathbb{Z} \setminus \{0\}.$$

Also, by definition of Y ,

$$S^k X \subseteq Y \quad \text{for } k = 1, \dots, 5.$$

Now pass to projective transformations. Since

$$S^6 = -I,$$

the projective class $\bar{S} \in \text{PGL}_6(\mathbb{R})$ has order 6. The projective class $\bar{T}$ has infinite order, because

$$T^m = I + m(T - I)$$

is not scalar for $m \neq 0$. The projective ping-pong lemma applied to

$$\langle \bar{T} \rangle \quad \text{and} \quad \langle \bar{S} \rangle$$

therefore gives

$$\langle \bar{T}, \bar{S} \rangle \cong \mathbb{Z} * \mathbb{Z} / 6\mathbb{Z}.$$

Lifting back from projective space, the only scalar matrices in Γ' are $\pm I$, and $-I = S^6$. Hence

$$\Gamma' \cong \langle T, -I \rangle *_{\langle -I \rangle} \langle S \rangle.$$

Since

$$\langle T, -I \rangle \cong \mathbb{Z} \times \mathbb{Z} / 2\mathbb{Z}$$

and

$$\langle S \rangle \cong \mathbb{Z} / 12\mathbb{Z},$$

we have

$$\Gamma' \cong (\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) *_{\mathbb{Z}/2\mathbb{Z}} \mathbb{Z}/12\mathbb{Z}.$$

5. THINNESS

The group

$$(\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) *_{\mathbb{Z}/2\mathbb{Z}} \mathbb{Z}/12\mathbb{Z}$$

is an amalgamated free product of amenable groups over a finite subgroup. Hence it has the Haagerup property.

Suppose, for contradiction, that Γ' had finite index in $\mathrm{Sp}_Q(\mathbb{Z})$. The group $\mathrm{Sp}_Q(\mathbb{Z})$ is an arithmetic lattice in the real Lie group $\mathrm{Sp}_6(\mathbb{R})$, which has real rank 3. Therefore $\mathrm{Sp}_Q(\mathbb{Z})$, and hence every finite-index subgroup of it, has Kazhdan's property (T). Thus Γ' would have property (T).

A discrete group with both the Haagerup property and property (T) is finite. But Γ' contains the infinite-order element T . This contradiction shows that

$$[\mathrm{Sp}_Q(\mathbb{Z}) : \Gamma'] = \infty.$$

Since Γ' is Zariski dense in Sp_Q , it is thin. By rational conjugacy, the original C-32 group

$$\Gamma = \langle A_0, B_0 \rangle$$

is thin in its natural arithmetic symplectic group.

The C-32 hypergeometric monodromy group is thin. More precisely,

$$\Gamma \cong (\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) *_{\mathbb{Z}/2\mathbb{Z}} \mathbb{Z}/12\mathbb{Z}$$

up to rational conjugacy, and Γ has infinite index in the integral points of its Zariski closure.

The parabolic branch $T^{-1}S$ fixes the projective point

$$q = [1 : -1 : 1 : -1 : 1 : -1].$$

This point lies on the boundary of K . Thus the cone certificate allows the ping-pong domain to accumulate on the boundary of the dominance chamber Δ_0 , which is precisely the feature that makes the C-32 case delicate.

APPENDIX A. THE MATRIX H

The matrix H is the 77×6 matrix whose rows are the following vectors.

h_0	h_1	h_2	h_3	h_4	h_5
1	-1	0	0	0	0
187	676	1105	1045	556	127
17	66	113	111	62	15
1	4	7	7	4	1
6	11	14	11	5	1
259	952	1573	1501	808	187
19	52	73	61	29	6
343	1276	2125	2041	1108	259
43	136	205	180	89	19
439	1648	2761	2665	1456	343
79	268	422	384	196	43
127	447	722	673	352	79
1	1	0	0	0	0
11	42	71	69	38	9
4	11	14	11	5	1
9	30	45	39	19	4
15	54	87	80	41	9
21	78	130	124	66	15
27	101	170	165	90	21
1	0	-1	0	0	0
5	19	32	31	17	4
1	0	1	0	0	0
1	0	0	-1	0	0
1	0	0	1	0	0
9	34	57	55	30	7
1	0	0	0	-1	0

h_0	h_1	h_2	h_3	h_4	h_5
4	15	25	24	13	3
1	0	0	0	1	0
1	0	0	0	0	-1
1	0	0	0	0	1
7	26	43	41	22	5
0	-5	-11	-14	-11	-5
4	-1	-5	-11	-14	-11
21	49	67	58	31	4
56	164	236	200	101	21
10	14	11	5	1	-5
36	99	135	109	55	10
81	261	395	341	170	36
10	5	-1	-5	-11	-14
45	111	145	121	64	10
114	350	509	431	215	45
220	745	1165	1039	525	114
13	11	5	1	-5	-11
54	138	181	148	76	13
132	413	608	518	257	54
4	1	-5	-11	-14	-11
19	49	67	58	31	4
46	142	208	178	91	19
88	298	466	415	211	46
4	11	14	11	5	-1
9	30	45	39	21	4
15	54	87	78	41	9
21	78	132	124	66	15
259	-187	-808	-1501	-1573	-952
19	-17	-70	-125	-127	-74
343	-259	-1108	-2041	-2125	-1276
43	-19	-89	-180	-205	-136
439	-343	-1456	-2665	-2761	-1648
79	-43	-196	-384	-422	-268
268	79	-43	-196	-384	-422
547	-439	-1852	-3373	-3481	-2068
127	-79	-352	-673	-722	-447
13	-11	-46	-83	-85	-50
9	-4	-19	-39	-45	-30
15	-9	-41	-80	-87	-54
54	15	-9	-41	-80	-87
36	-10	-55	-109	-135	-99
81	-36	-170	-341	-395	-261
496	144	-81	-369	-721	-793
114	-45	-215	-431	-509	-350
132	-54	-257	-518	-608	-413
88	-46	-211	-415	-466	-298
9	-4	-21	-39	-45	-30
15	-9	-41	-78	-87	-54
54	15	-9	-41	-78	-87
21	-15	-66	-124	-132	-78
78	21	-15	-66	-124	-132

APPENDIX B. EXACT VERIFICATION

The following SageMath code verifies the finite cone certificate and the Lie-algebra computation. It uses only integer and rational arithmetic. To run it, set `H_data` equal to the rows displayed in Appendix A.

```
from sage.all import *
```

```
H_data = [
[1,-1,0,0,0,0],
```

```

[187,676,1105,1045,556,127],
[17,66,113,111,62,15],
[1,4,7,7,4,1],
[6,11,14,11,5,1],
[259,952,1573,1501,808,187],
[19,52,73,61,29,6],
[343,1276,2125,2041,1108,259],
[43,136,205,180,89,19],
[439,1648,2761,2665,1456,343],
[79,268,422,384,196,43],
[127,447,722,673,352,79],
[1,1,0,0,0,0],
[11,42,71,69,38,9],
[4,11,14,11,5,1],
[9,30,45,39,19,4],
[15,54,87,80,41,9],
[21,78,130,124,66,15],
[27,101,170,165,90,21],
[1,0,-1,0,0,0],
[5,19,32,31,17,4],
[1,0,1,0,0,0],
[1,0,0,-1,0,0],
[1,0,0,1,0,0],
[9,34,57,55,30,7],
[1,0,0,0,-1,0],
[4,15,25,24,13,3],
[1,0,0,0,1,0],
[1,0,0,0,0,-1],
[1,0,0,0,0,1],
[7,26,43,41,22,5],
[0,-5,-11,-14,-11,-5],
[4,-1,-5,-11,-14,-11],
[21,49,67,58,31,4],
[56,164,236,200,101,21],
[10,14,11,5,1,-5],
[36,99,135,109,55,10],
[81,261,395,341,170,36],
[10,5,-1,-5,-11,-14],
[45,111,145,121,64,10],
[114,350,509,431,215,45],
[220,745,1165,1039,525,114],
[13,11,5,1,-5,-11],
[54,138,181,148,76,13],
[132,413,608,518,257,54],
[4,1,-5,-11,-14,-11],
[19,49,67,58,31,4],
[46,142,208,178,91,19],
[88,298,466,415,211,46],
[4,11,14,11,5,-1],
[9,30,45,39,21,4],
[15,54,87,78,41,9],
[21,78,132,124,66,15],
[259,-187,-808,-1501,-1573,-952],
[19,-17,-70,-125,-127,-74],
[343,-259,-1108,-2041,-2125,-1276],
[43,-19,-89,-180,-205,-136],
[439,-343,-1456,-2665,-2761,-1648],
[79,-43,-196,-384,-422,-268],
[268,79,-43,-196,-384,-422],
[547,-439,-1852,-3373,-3481,-2068],
[127,-79,-352,-673,-722,-447],
[13,-11,-46,-83,-85,-50],
[9,-4,-19,-39,-45,-30],
[15,-9,-41,-80,-87,-54],
[54,15,-9,-41,-80,-87],
[36,-10,-55,-109,-135,-99],
[81,-36,-170,-341,-395,-261],
[496,144,-81,-369,-721,-793],
[114,-45,-215,-431,-509,-350],
[132,-54,-257,-518,-608,-413],
[88,-46,-211,-415,-466,-298],
[9,-4,-21,-39,-45,-30],
[15,-9,-41,-78,-87,-54],
[54,15,-9,-41,-78,-87],
[21,-15,-66,-124,-132,-78],
[78,21,-15,-66,-124,-132]
]

```



```

H = matrix(ZZ, H_data)

S = matrix(ZZ, [
  [0,0,0,0,0,1],
  [-1,0,0,0,0,0],
  [0,-1,0,0,0,0],
  [0,0,-1,0,0,0],
  [0,0,0,-1,0,0],
  [0,0,0,0,-1,0]
])

T = matrix(ZZ, [
  [1,5,11,14,11,5],
  [0,1,0,0,0,0],
  [0,0,1,0,0,0],
  [0,0,0,1,0,0],
  [0,0,0,0,1,0],
  [0,0,0,0,0,1]
])

E = matrix(ZZ, [
  [1,0,0,0,0,0],
  [0,0,0,0,0,-1],
  [0,0,0,0,-1,0],
  [0,0,0,-1,0,0],
  [0,0,-1,0,0,0],
  [0,-1,0,0,0,0]
])

I6 = identity_matrix(ZZ,6)

assert S^6 == -I6
assert S^12 == I6
assert E^2 == I6
assert E*S*E == S.inverse()
assert E*T*E == T.inverse()

Q = matrix(ZZ, [
  [0,5,11,14,11,5],
  [-5,0,5,11,14,11],
  [-11,-5,0,5,11,14],
  [-14,-11,-5,0,5,11],
  [-11,-14,-11,-5,0,5],
  [-5,-11,-14,-11,-5,0]
])

assert S.transpose()*Q*S == Q
assert T.transpose()*Q*T == Q
assert Q.det() != 0

# Cone interior check.
z = vector(ZZ, [47,-1,-1,-1,-1,-1])
assert min((H*z).list()) > 0

# The seed rays are not used in the proof, but this confirms solidity.
R = matrix(ZZ, [
  [1,5,11,14,11,5],
  [0,-1,0,0,0,0],
  [0,0,-1,0,0,0],
  [0,0,0,-1,0,0],
  [0,0,0,0,-1,0],
  [0,0,0,0,0,-1]
])
assert R.det() != 0
assert min((H*R).list()) >= 0

# Branch maps.
G = [T.inverse()]
for k in range(1,6):
    eps = (-1)^(k+1)
    G.append(eps*T.inverse()*(S^k))
    G.append(eps*T.inverse()*(S^k)*E)

# Dual cone generated by the rows of H.
row_cone = Cone([vector(QQ, r) for r in H.rows()])

```

```

# Exact cone-containment checks:
# G_i K subset K iff each row h*G_i is in the row cone of H.
for Gi in G:
    for h in H.rows():
        assert row_cone.contains(vector(QQ, h*Gi))

# Lie algebra generated by the conjugates of N=T-I.
def flat(M):
    return vector(QQ, M.list())

def rank_of(mats):
    if len(mats) == 0:
        return 0
    return matrix(QQ, [flat(M) for M in mats]).rank()

basis = []

def add_to_basis(M):
    global basis
    r0 = rank_of(basis)
    r1 = rank_of(basis + [M])
    if r1 > r0:
        basis.append(M)
        return True
    return False

N = T - I6
for i in range(6):
    add_to_basis((S^i)*N*(S^(-i)))

changed = True
while changed:
    changed = False
    current = list(basis)
    for X in current:
        for Y in current:
            if add_to_basis(X*Y - Y*X):
                changed = True

assert len(basis) == 21
for X in basis:
    assert X.transpose()*Q + Q*X == 0

print("All C-32 checks passed exactly over QQ.")

```

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